Ex: 7
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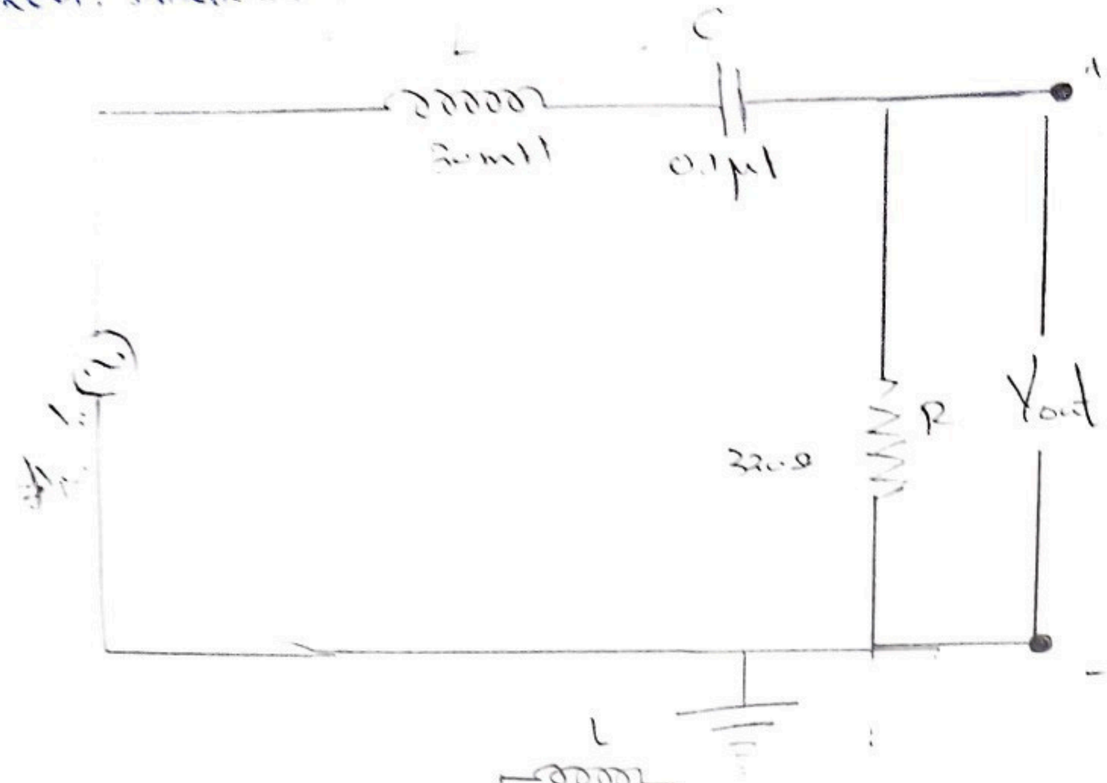
FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS.

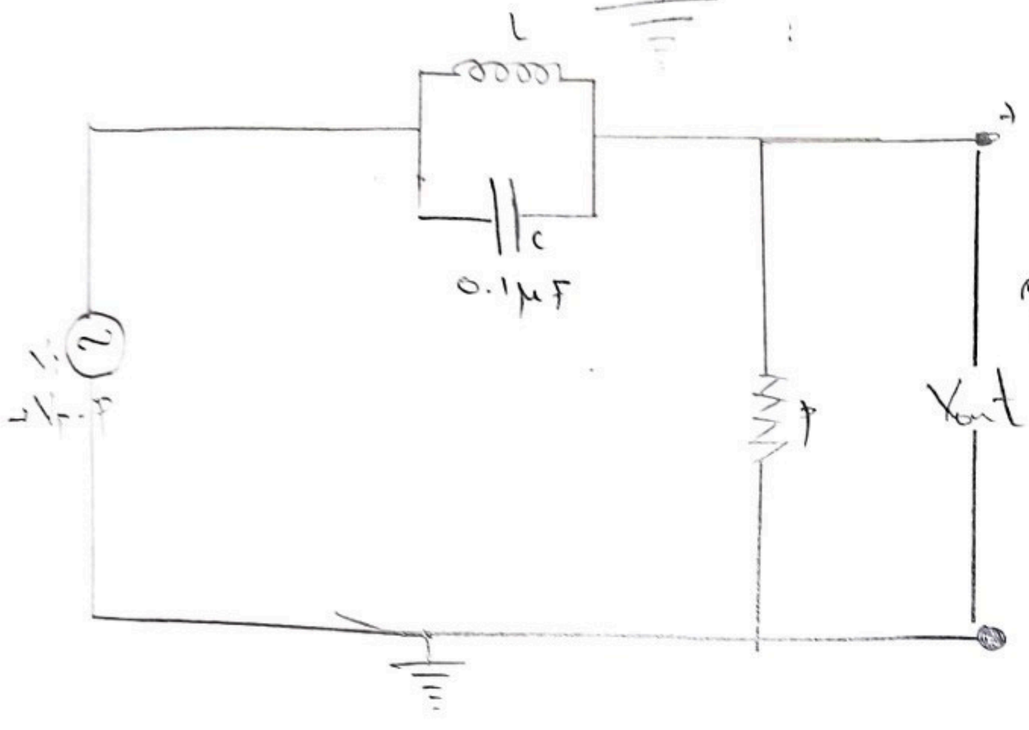
AIM:

To study the behaviour of series and parallel RLC Circuits at resonance and to determine the resonant frequency, bandwidth and -2 of the given RLC Circuit using Multisim Simulator.

APPARATUS REQUIRED.

COMPONENTS	SPECIFICATIONS	QUANTITY.
CRO	10MHz	1
DIB	–	1
Capacitors	0.1μF	1
Resistors	330Ω	1
Bread board	–	–
Connecting wires	–	–
Pc with Multisim simulator.	–	–

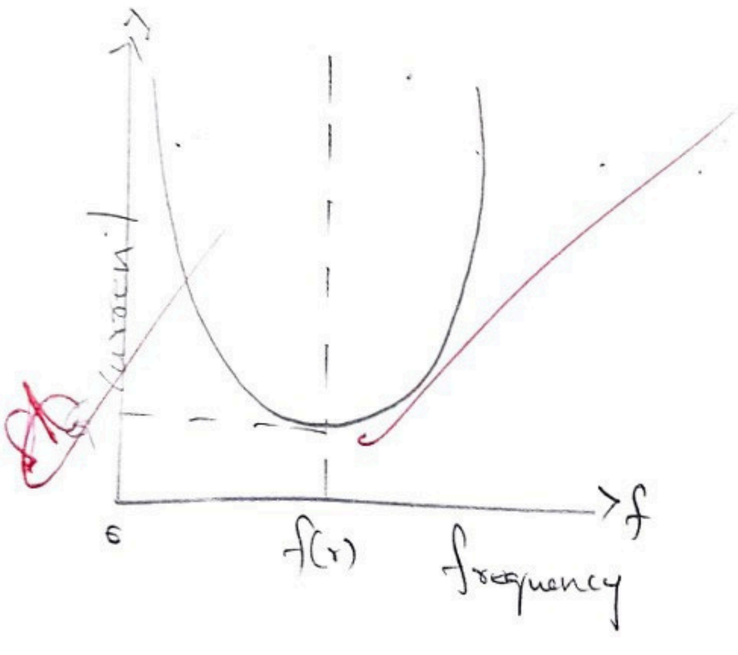


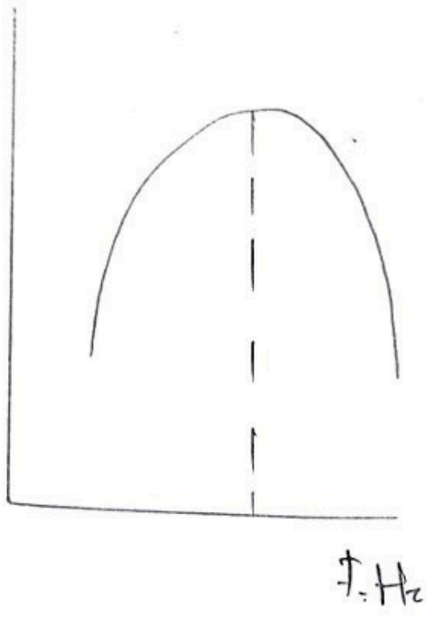


Parallel RLC Circuit.

Parallel Resonance

Series Resonance.





FORMULA:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW = f_2 - f_1$$

$$Q = f_0 / BW$$

PROCEDURE:

- Construct the circuit as shown in the given figure on the bread board.
- Connect the CRO - CH1 to input and CH2 to output
- Set the input voltage to about 4V Peak-to-Peak.
- Vary the frequency from 5000 Hz to 3kHz and observe the output voltage to find the maximum output
- Find the cut off frequency where $V_{out} = \frac{V_m}{\sqrt{2}}$, then calculate bandwidth and Q-factor.

CALCULATION:

$L = 100\,mH = 0.1\,H$

$C = 0.25\,\mu F = 0.25 \times 10^{-6} F$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25\times10^{-6})}}$$

$$f_0 = \frac{1}{2\pi\sqrt{1.58\times10^{-4}}}$$

$$f_0 = \frac{1}{9.92\times10^{-4}}$$

$$f_0 = 1000\,Hz$$

Maximum Current (Series) and minimum Current (parallel) Occurs at 1000Hz.

TABULATION :

$V_{in} = 6V_{p-p}$

SERIES RESONANCE			PARALLEL RESONANCE		
Frequency (Hz)	output Voltage (V)	output Current (mA)	Frequency (Hz)	output Voltage (V)	output Current (mA)
500	1.2	2	500	1.5	9
700	2.0	4	700	2.8	7
900	3.5	7	900	4.0	5
(Resonance) 1000	5.0	10	(Resonance) 1000	5.0	2 (minimum)
1100	3.6	7	1100	4.1	5

Not required

Op

MARKS SPLIT-UP:

DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
Experimental Setup / Connections	20	20
observation / Readings	20	20
Calculations / Results	20	20
Viva	20	20
Record	20	20
	Total	100

100

1/3

RESULT:

Thus the phenomenon of resonance in RLC Circuit was studied ~~of~~ the following was determined using multisim Simulator.

Series RLC Circuit:

Resonant frequency=

Bandwidth=

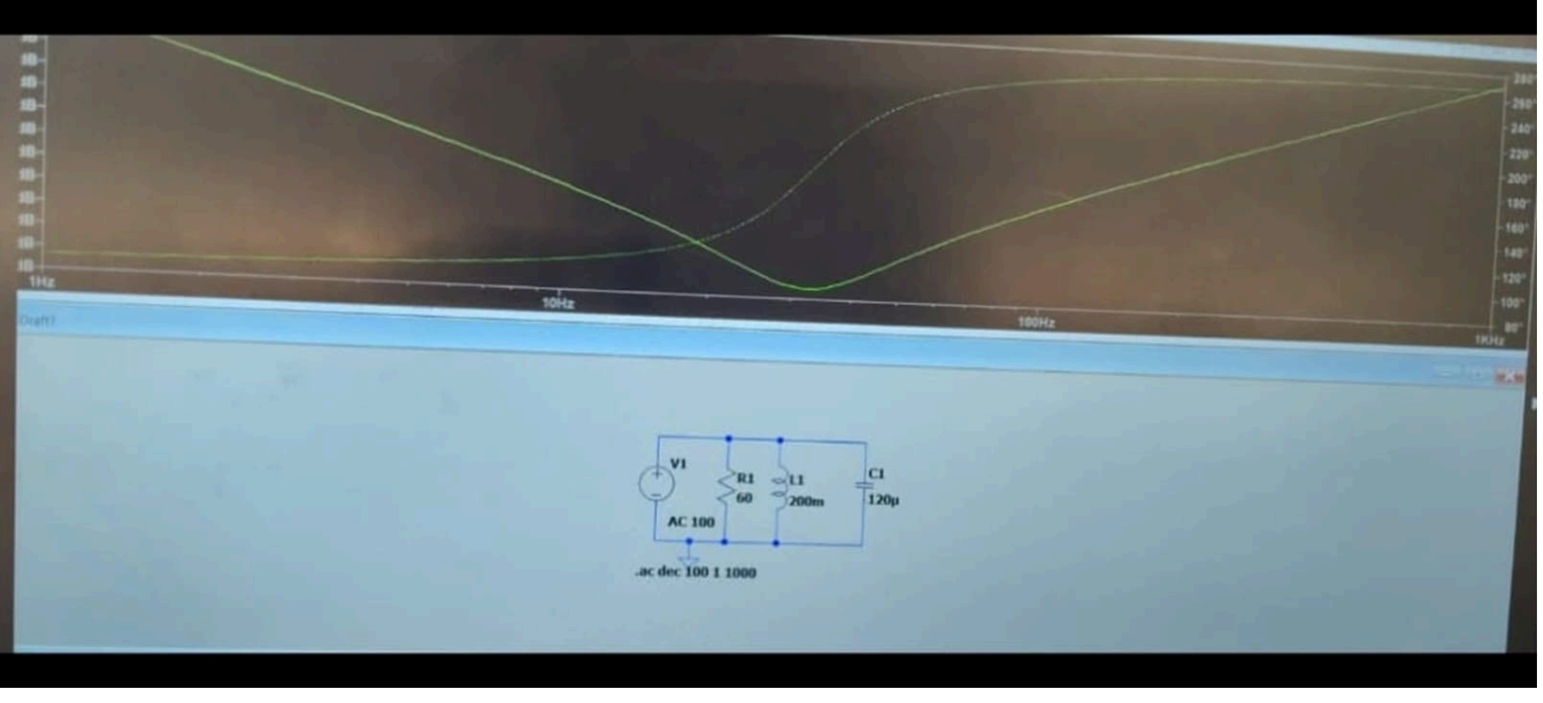
Q factor=

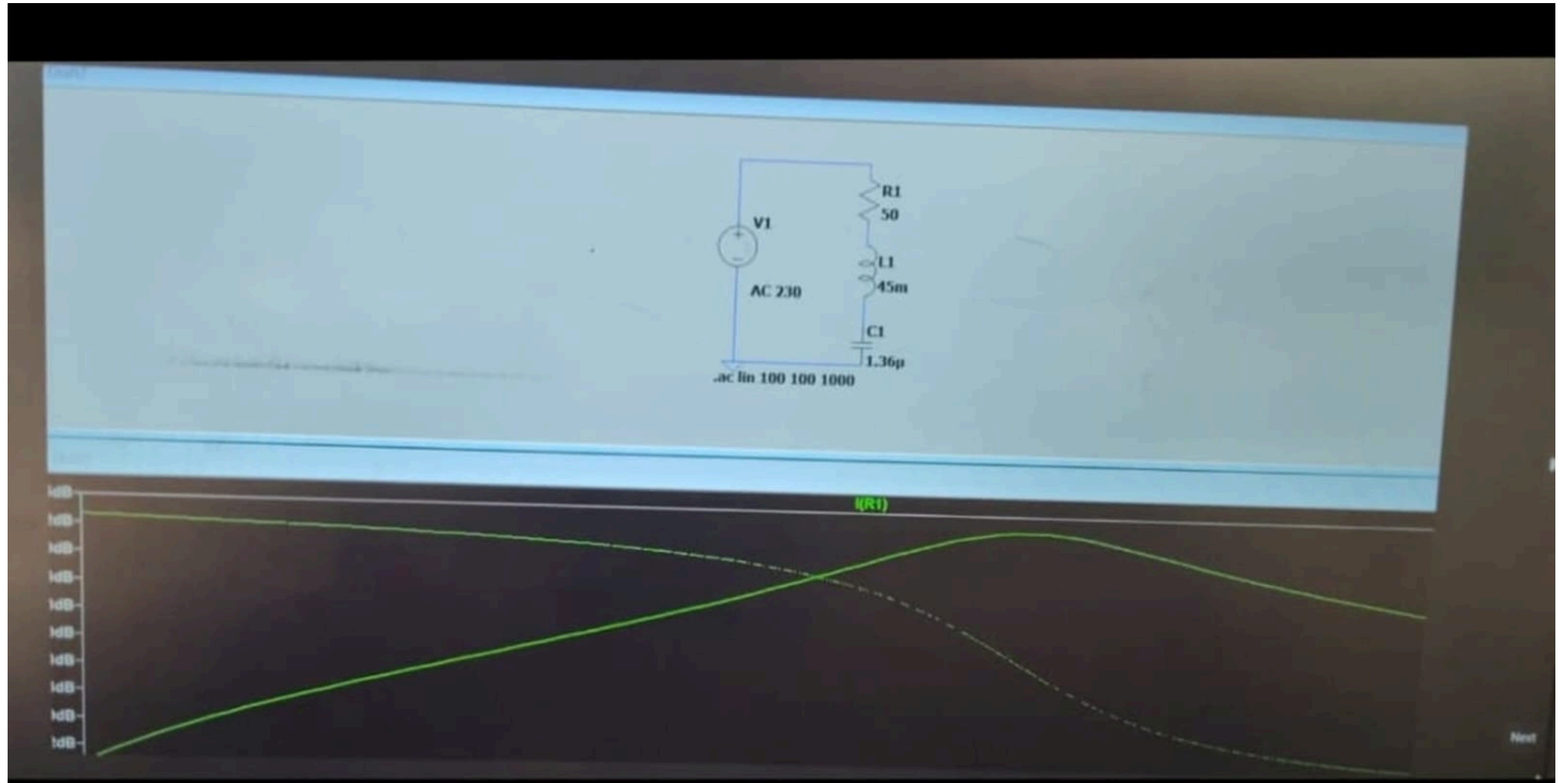
Parallel RLC Circuit:

Resonant frequency=

Bandwidth=

Q factor=


1Hz
10Hz
100Hz
1KHz
V1
AC 100
R1
60
L1
200m
C1
120µ
.ac dec 100 1 1000


V1
AC 230
R1
50
L1
45m
C1
1.36µ
.ac lin 100 100 1000
I(R1)